

HCNI6222 2026 FT DISD0601 TERM 2 GRP 4

Group 04

ST: 1049 2868

CampusConnect is a website prototype designed to help college students discover and register for campus activities through a single platform. The purpose of the project is to improve access to information about academic, sports, and social events while encouraging student participation and interaction. The website was developed using HTML, CSS, and JavaScript, with a user-friendly interface that includes event categories, a search bar, filtering options, event details, registration functionality, and a My Events section.

HLULANI TYRONE MTHOMBENI
Human Computer Interaction

Table of contents

Contents

1.1 Website Functionality 2
1.2 Design of the Website 3
1.2.1 Usability Goals 3
1.2.2 Desirable User Experience 4
1.2.3 Design Principles 4
1.2.4 Interaction Types 4
1.2.5 Social Interactions 5
1.2.6 Emotional Interaction 5
1.2.7 WCAG 2.0 Accessibility Guidelines 6
2.1 Introduction 6
2.2 Technologies used 7
2.3 Website Structure and Navigation 7
2.4 Interactive Features and Functionality 8
4.1 Usability Goals 9
4.2 Desirable User Experience 10
4.3 Design Principles 11
4.5 Social Interactions 12
4.6 Emotional Interaction 13
4.7 WCAG 2.0 Accessibility Guidelines 14
References 16

1.1 Website Functionality

Website Name: **CampusConnect**

CampusConnect is a website designed to help college students discover campus activities and events. The purpose of the website is to make it easier for students to find academic workshops, sports activities, and social events in one convenient place.

The website addresses the problem of students missing campus activities because event information may be difficult to find or spread across different communication platforms.

Target Users:

The primary users are college students who want to participate in campus activities, improve their academic skills, engage in sports, and connect with other students.

Main Features

- **Homepage:** Introduces CampusConnect and displays images and information about campus activities.
- **Event Categories:** Organises activities into Academic, Sports, and Social categories.
- **Search Function:** Allows students to search for events using keywords.
- **Category Filtering:** Enables students to display events from a selected category.
- **Event Details:** Displays event descriptions, dates, and locations through a popup window.
- **Event Registration:** Allows students to register for events they are interested in.
- **My Events:** Displays the events a student has registered for.
- **Cancel Registration:** Allows students to cancel an event registration.
- **Saved Registrations:** Uses browser local storage to keep registrations available after refreshing the page.

Usefulness to Students

CampusConnect provides a central place where students can explore activities, find event information, and manage their registrations. Its search and filtering features

reduce the effort required to find relevant events. The website also encourages students to participate in campus life, develop academic skills, and build social connections.

1.2 Design of the Website

The CampusConnect website is designed using Human-Computer Interaction (HCI) principles to provide students with a simple, attractive, and user-friendly experience. The website uses a dark purple color scheme with white text and light purple accents to create a consistent visual identity. The layout includes a navigation bar, event categories, search and filtering features, event cards, and registration functionality.

The design focuses on making information easy to find, reducing the number of steps required to complete tasks, and helping students interact with the website comfortably. The website is also designed to support different screen sizes and provide clear feedback when users perform actions.

1.2.1 Usability Goals

The CampusConnect website focuses on the following usability goals:

Effectiveness: Students can find event information, view event details, register for activities, and manage their registrations.

Efficiency: The search bar and category filters help students locate relevant events without browsing through every event card.

Safety: Confirmation and status messages help students understand whether their registration has been added or cancelled. Users can also review their registered events before cancelling the registration.

Utility: The website provides useful features, including event browsing, event details, registration, search, filtering, and the My Events section.

Learnability: The website uses familiar navigation links, clearly labelled buttons, and organized event categories so that new users can understand how to use it.

Memorability: The consistent layout, colors, navigation, and button styles help returning users remember how to find and manage events.

1.2.2 Desirable User Experience

CampusConnect aims to provide positive user experience by making campus activities easy to discover and explore. The website uses an organized layout, readable text, clear headings, and visually distinct event cards.

Students receive feedback after registering for or cancelling an event, helping them understand the result of their actions. The event that details popup also allows users to view an event's date, location, and description without navigating to another page.

The website aims to make students feel informed, confident, and encouraged to participate in academic, sports, and social activities.

1.2.3 Design Principles

The following HCI design principles are applied to CampusConnect:

Visibility: The navigation bar, search field, category filters, and event buttons are visible and easy to locate.

Consistency: The website uses consistent colors, fonts, spacing, event cards, and button styles across its sections.

Feedback: Success messages inform users when they register for or cancel an event.

Affordance: Buttons are styled to look clickable, helping users recognize available actions.

Mapping: Navigation links correspond to website sections, while event registration buttons are placed alongside event information.

Constraints: Category filters limit the displayed events to a selected category, helping users focus on relevant information.

Simplicity: The website groups activities into Academic, Sports, and social categories to reduce unnecessary complexity.

These principles help make the website easier to understand and use.

1.2.4 Interaction Types

CampusConnect uses several interaction types:

Instructing: Users click navigation links, select a category filter, or press a button to perform an action.

Conversing: The website communicates with users through registration messages and event status information.

Manipulating: Users interact directly with event cards, search results, category filters, and the event details popup.

Exploring: Users browse the website's different categories to discover activities that interest them.

These interaction types provide different ways for students to access information and complete tasks.

1.2.5 Social Interactions

CampusConnect supports social engagement by helping students discover activities where they can meet other students and participate in campus life.

The Social category includes events intended to encourage communication, friendship, and interaction. Academic events also provide opportunities for students to participate in study groups and collaborative workshops, while sports activities encourage teamwork.

Although the current prototype does not include built-in chat, comments, or direct messaging, it helps students discover events that may create opportunities for face-to-face social interaction.

1.2.6 Emotional Interaction

The design of CampusConnect aims to create a welcoming and positive experience. The purple color scheme, organized layout, event imagery, and rounded cards contribute to a modern and visually consistent appearance.

Clear registration messages help users feel confident that their actions have been processed. The My Events section also helps users keep track of their selected activities.

By making event information easier to find, the website aims to reduce frustration and encourage students to become more involved in campus activities.

1.2.7 WCAG 2.0 Accessibility Guidelines

CampusConnect considers the Web Content Accessibility Guidelines (WCAG) 2.0, which aims to make websites more accessible to people with different abilities.

Perceivable: Text, headings, and event information should be easy to see and read. Images should include meaningful alternative text, and text should have sufficient contrast against its background.

Operable: Navigation links, buttons, search controls, and popup functions should be usable with a keyboard as well as a mouse. Keyboard focus should be visible.

Understandable: The website uses clear labels, familiar navigation, consistent layouts, and understandable messages to help users know what actions are available and what happens after they use them.

Robust: The website should use valid, semantic HTML and standard browser functionality so that it works reliably across supported browsers and can be interpreted by assistive technologies.

These four principles—Perceivable, Operable, Understandable, and Robust—form the foundation of WCAG 2.0. The website should be tested to verify that these accessibility requirements are met.

2.1 Introduction

The CampusConnect website prototype was developed to help college students discover, explore, and register for campus activities through a single platform. The website focuses on improving students' access to information about academic, sports, and social events.

The prototype was developed using HTML, CSS, and JavaScript. HTML was used to structure the website, CSS was used to create the visual design and layout, and JavaScript was used to implement interactive features such as searching, filtering, viewing event details, registering for events, and managing registered activities.

The design aims to provide user-friendly experience through clear navigation, readable text, consistent styling, and feedback messages. The prototype allows students to explore available activities and manage their registrations through the My Events section.

2.2 Technologies used

The following technologies were used to develop the CampusConnect prototype:

- **HTML:** Structures the website content, including the navigation menu, event cards, search bar, and registration sections.
- **CSS:** Controls the website's appearance, including the dark purple color scheme, typography, spacing, responsive layout, and interactive visual effects.
- **JavaScript:** Provides functionality for searching and filtering events, displaying event details, registering for activities, cancelling registrations, and updating the My Events section.
- **Browser local Storage:** Stores registered events locally in the user's browser so that registrations can remain available when the page is refreshed in the same browser.

These technologies were selected because they support the development of an interactive website prototype without requiring a backend server or database.

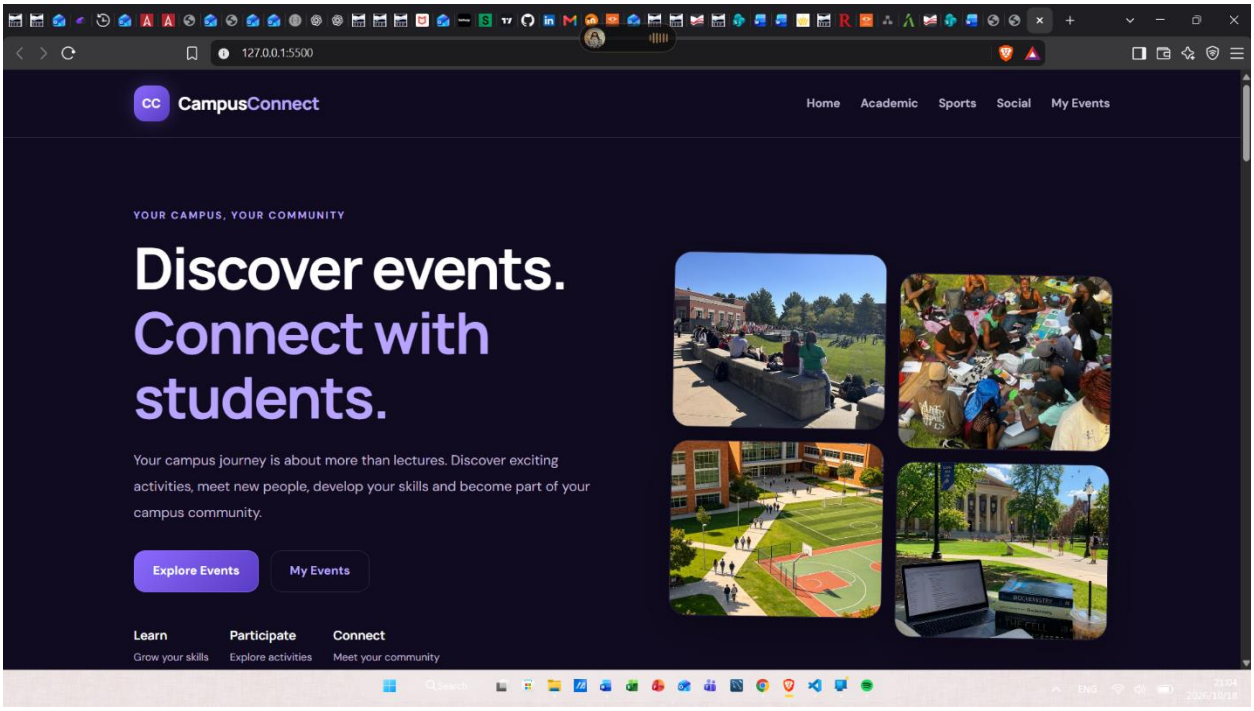
2.3 Website Structure and Navigation

The CampusConnect website uses a simple navigation structure to help students access different sections of the website. The navigation menu contains Home, Academic, Sports, Social, and My Events. Each section serves a specific purpose and allows users to explore activities according to their interests.

The Home section introduces the website and displays event images to attract users' attention. The Academic section displays learning-related activities, such as study group meetups and academic workshops. The Sports section presents sporting activities, while the social section provides opportunities for students to discover social events and connect with others.

The My Events section allows users to view the activities they have registered for. The navigation structure is designed to reduce confusion by keeping the main sections clearly labelled and accessible.

Overall, the website structure supports ease of navigation and allows students to move between event categories without needing to search through unrelated information.



2.4 Interactive Features and Functionality

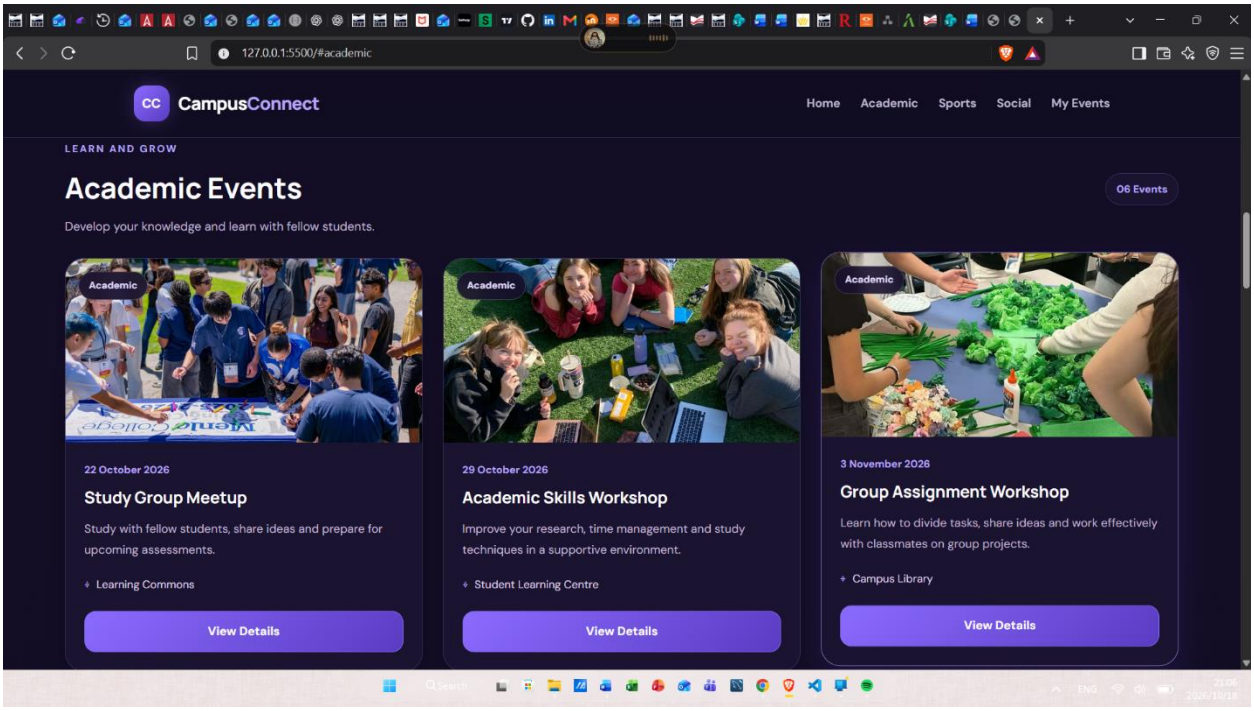
The CampusConnect prototype includes interactive features that allow students to explore campus activities and manage their event registrations. JavaScript was used to make the website respond to user actions and provide a more engaging experience.

The search function allows students to find events by entering keywords. The category filter enables students to display academic, sports, or social activities according to their interests. These features make it easier to locate relevant events without browsing through every event card.

The View Details feature allows students to open a popup containing additional information about a selected event, including its date, location, and description. Students can then register for an event or cancel a registration when necessary.

The My Events section displays the events selected by the user. Registration and cancellation messages provide feedback so that users can understand whether their actions have been completed. The prototype also uses browser local Storage to retain registration information when the page is refreshed in the same browser.

These interactive features support usability by helping students find information, make decisions, and manage their selected activities more efficiently.



4.1 Usability Goals

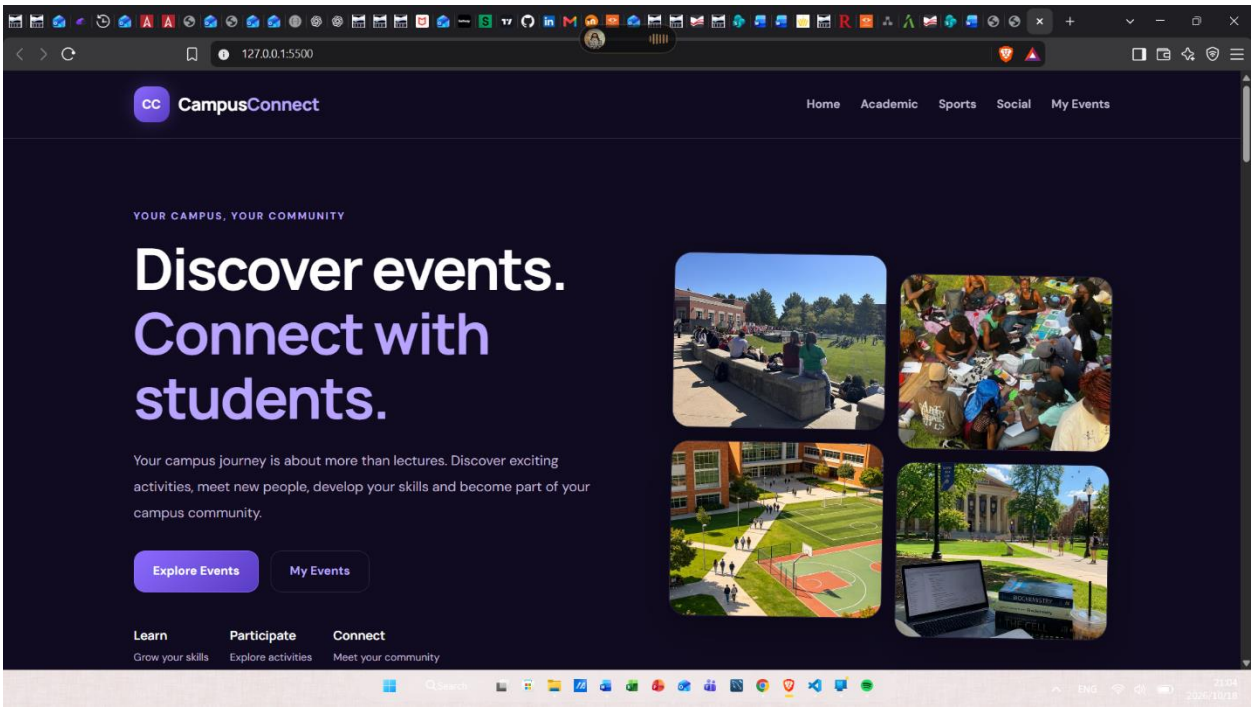
You can use this explanation in your research report:

Usability Goals of CampusConnect

CampusConnect was designed to help college students discover and register for campus activities in a simple and efficient way. The prototype focuses on effectiveness, efficiency, safety, utility, learnability, and memorability.

- Effectiveness: Students can view event details, register for events, and manage their registrations through the My Events section.
- Efficiency: The search bar and category filters allow students to find relevant events quickly.
- Safety: Registration status messages inform users about their actions, while the cancellation option allows them to manage their registrations.
- Utility: The website provides useful features such as event categories, event descriptions, search, filtering, and registration.
- Learnability: Clear navigation links, familiar buttons, and organized event cards help new users understand how to use the website.

- Memorability: Consistent colors, layouts, navigation, and button styles help returning users remember how to use the website.



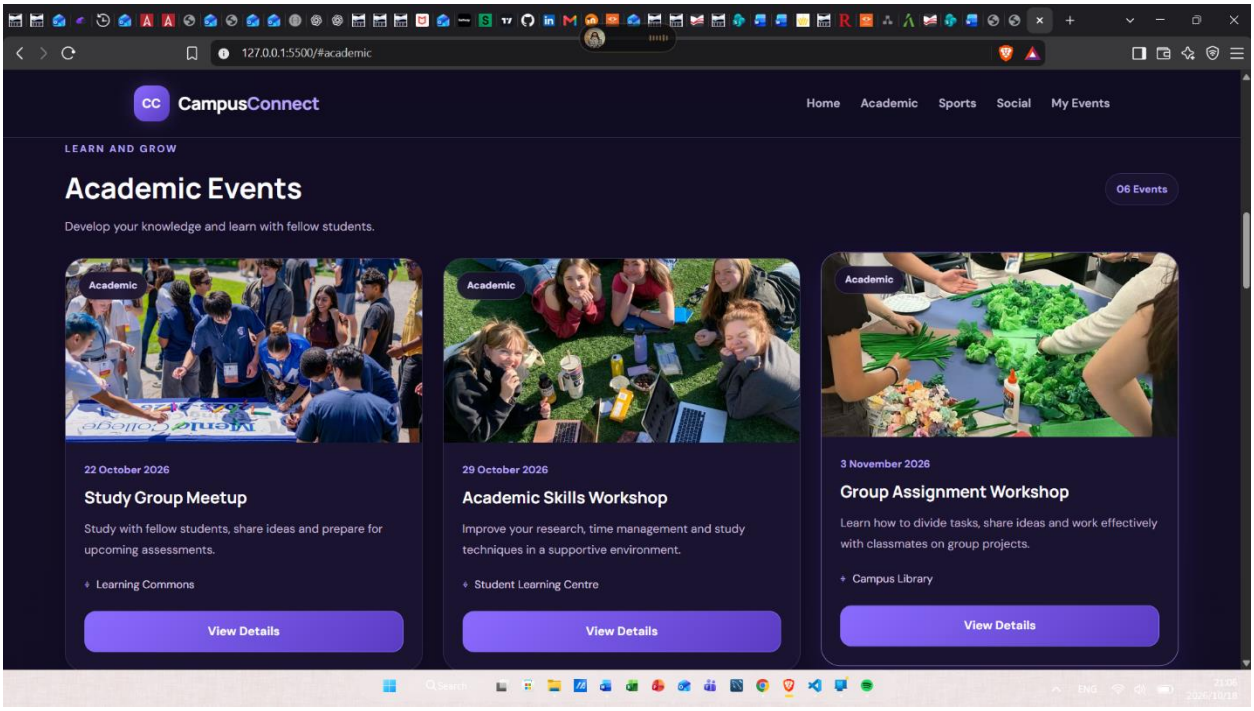
4.2 Desirable User Experience

The CampusConnect aims to provide students with positive, engaging, and user-friendly experience when discovering campus activities. The website uses a dark purple color scheme, clear headings, readable text, and organized event cards to create a consistent visual appearance.

The navigation menu allows students to move between the Home, Academic, Sports, Social, and My Events sections. The search bar and category filters help users find relevant activities without browsing through every event. When students select an event, a popup displays its title, date, location, and description.

The website also provides feedback when users register for or cancel an event. The My Events section allows students to view their registered activities, helping them keep track of their choices.

These features aim to make the website convenient to use, reduce frustration, and encourage students to participate in academic, sports, and social activities.



4.3 Design Principles

The CampusConnect website applies Human-Computer Interaction design principles to make the interface clear, consistent, and easy to use. These principles help students understand the available features and interact with the website effectively.

Visibility: The navigation menu, search bar, category filters, and event buttons are displayed so users can identify the main features of the website.

Consistency: The website uses a consistent dark purple color scheme, typography, button styles, and event card layouts across its sections. This creates a unified appearance and helps users become familiar with the interface.

Feedback: When users register for or cancel an event, the website displays a message indicating the outcome of their action. The registration status is also updated in the interface.

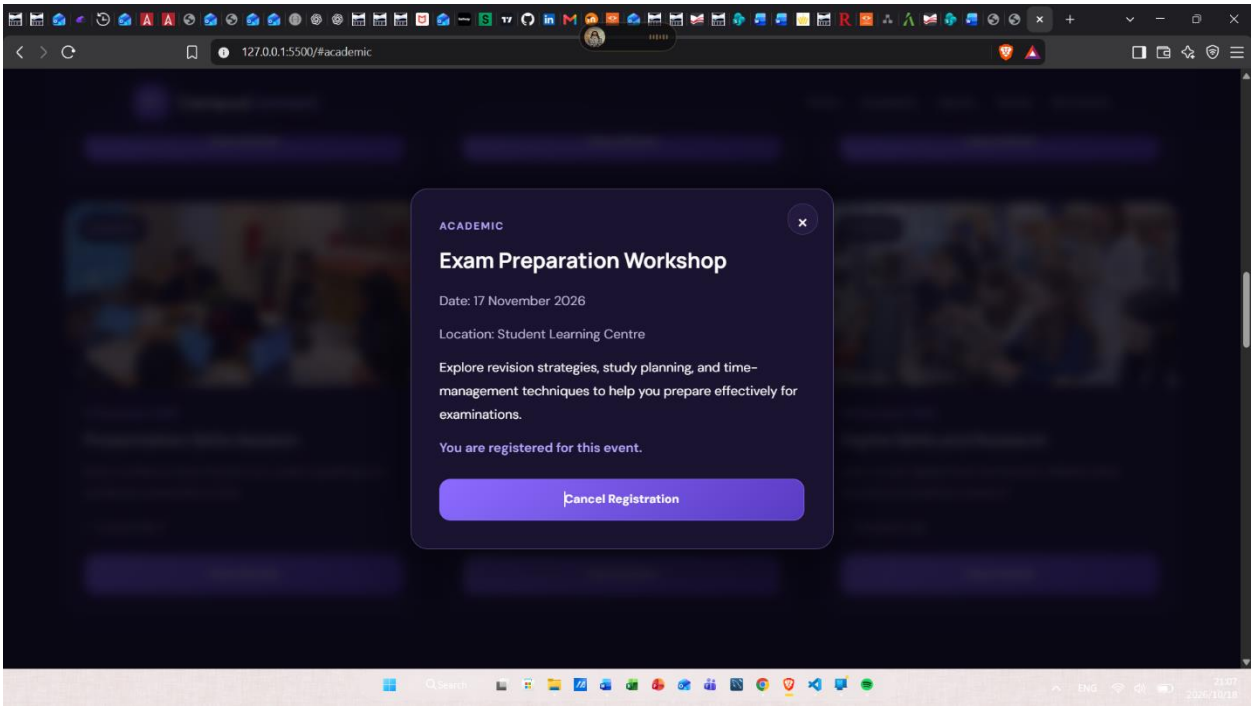
Affordance: Buttons such as "View Details" and "Register for Event" are styled to look interactive, helping users recognize the actions they can perform.

Mapping: Navigation links correspond to the sections they open. For example, selecting "Sports" takes users to the sports events section, while "My Events" provides access to their registered activities.

Constraints: The category filter allows users to narrow the displayed events to Academic, Sports, or social activities. This reduces unnecessary information and helps users focus on relevant events.

Simplicity: Events are organized into categories, and each event card presents essential information. This helps users browse activities without dealing with an unnecessarily complicated interface.

Overall, these design principles support the usability of CampusConnect by making its features easier to recognize, understand, and operate.



4.5 Social Interactions

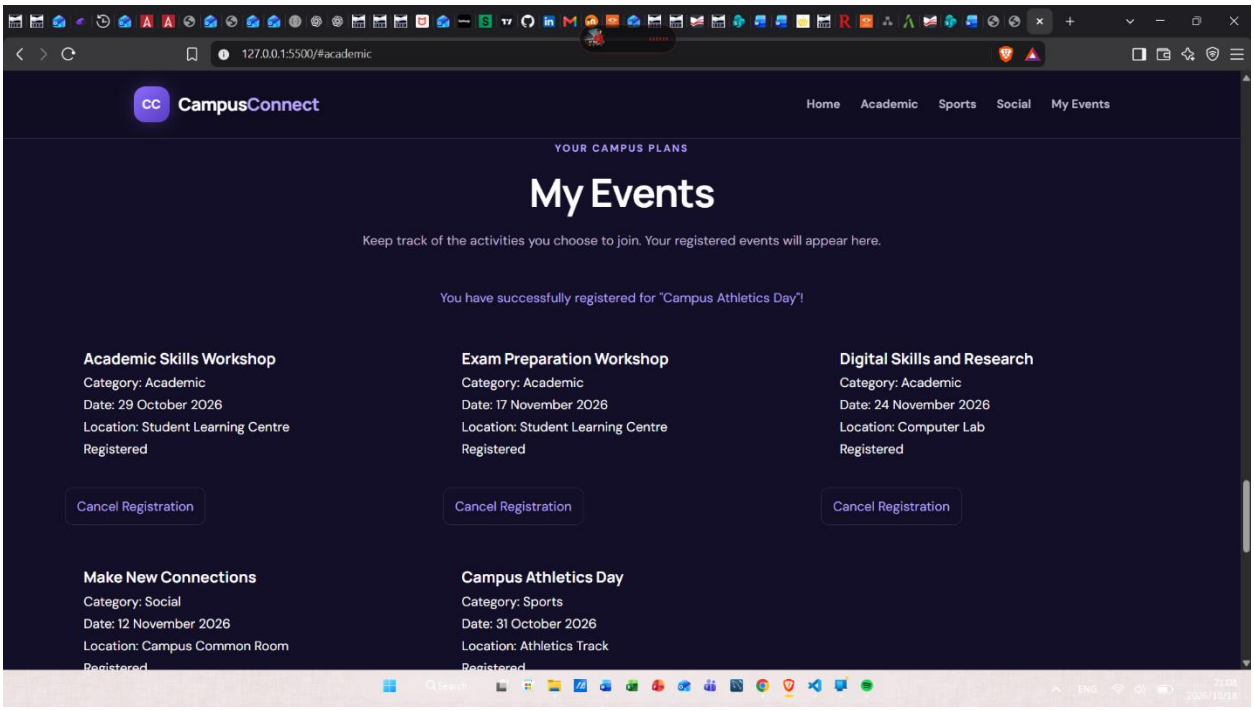
Social interaction is an important part of the CampusConnect website because it helps students discover activities where they can meet other students, share experiences, and develop connections within the college community.

The website includes Academic, Sports, and social event categories. Academic activities, such as study group meetups and group assignment workshops, provide opportunities for students to learn collaboratively and exchange ideas. Sports activities encourage teamwork and participation, while social events give students opportunities to meet people from different courses and build friendships.

The event details popup displays information about each activity, including its date, location, and description. This helps students understand what an event involves before deciding whether to register. The My Events section also allows students to keep track of the activities they have selected.

Although the current CampusConnect prototype does not include built-in chat, direct messaging, or discussion forums, it supports social engagement by helping students discover events that encourage face-to-face interaction.

Overall, CampusConnect aims to strengthen student participation and create opportunities for collaboration, communication, and social connection within the college environment.



4.6 Emotional Interaction

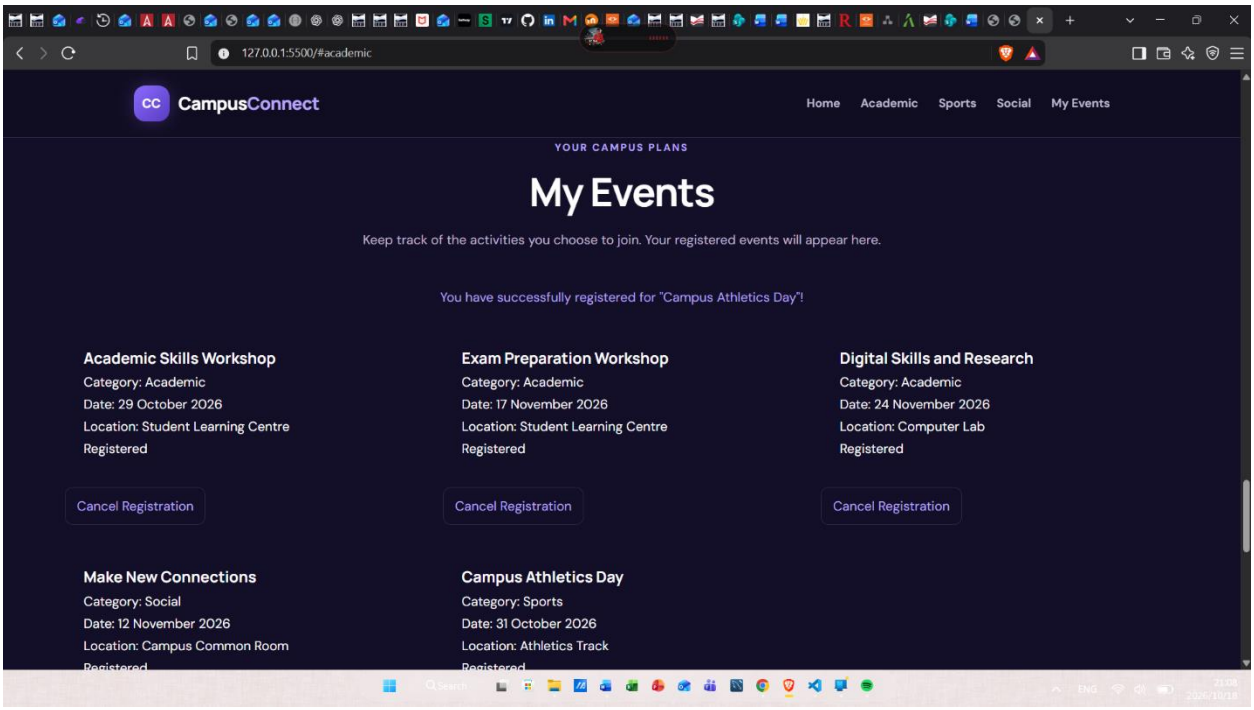
Emotional interaction focuses on how the design and functionality of a website influence users' feelings and experiences. CampusConnect aims to create a welcoming, comfortable, and positive experience for college students when they explore campus activities.

The website uses a dark purple color scheme with light purple accents and white text to create a consistent visual identity. Its organized layout, event cards, and clear headings help present information in a structured and approachable way.

CampusConnect also provides feedback when users register for or cancel an event. These messages help users understand the outcome of their actions and can increase their confidence when using the website. The My Events section allows students to review their registered activities, helping them feel more organized.

The website encourages students to explore academic, sports, and social activities that may help them develop new skills, work with other students, and form social connections. Simple navigation and search functionality are intended to reduce frustration when looking for events.

Overall, CampusConnect aims to promote confidence, convenience, and a sense of connection by providing students with an accessible way to discover and manage campus activities.



4.7 WCAG 2.0 Accessibility Guidelines

The CampusConnect website considers the Web Content Accessibility Guidelines (WCAG) 2.0 to help make the website usable by people with different abilities. WCAG 2.0 is based on four main principles: Perceivable, Operable, Understandable, and Robust (POUR).

1. Perceivable

Information and interface elements should be presented in ways users can perceive. CampusConnect uses white text and light purple accents against a dark background to distinguish content and interface elements. Event headings and descriptions organize information clearly. To improve accessibility, images should have meaningful alternative text, and text and interactive elements should meet the recommended contrast requirements.

2. Operable

Users should be able to navigate and operate the website using different input methods. CampusConnect provides navigation links, search controls, category filters, and buttons for viewing event details and registering. Keyboard navigation, visible focus indicators, and the ability to close the event popup using the keyboard should be tested to verify that the website is operable without a mouse.

3. Understandable

Website content and controls should be clear and predictable. CampusConnect uses familiar navigation labels, organized event categories, and descriptive button text. Feedback messages inform users about registration actions, helping them understand the system's response.

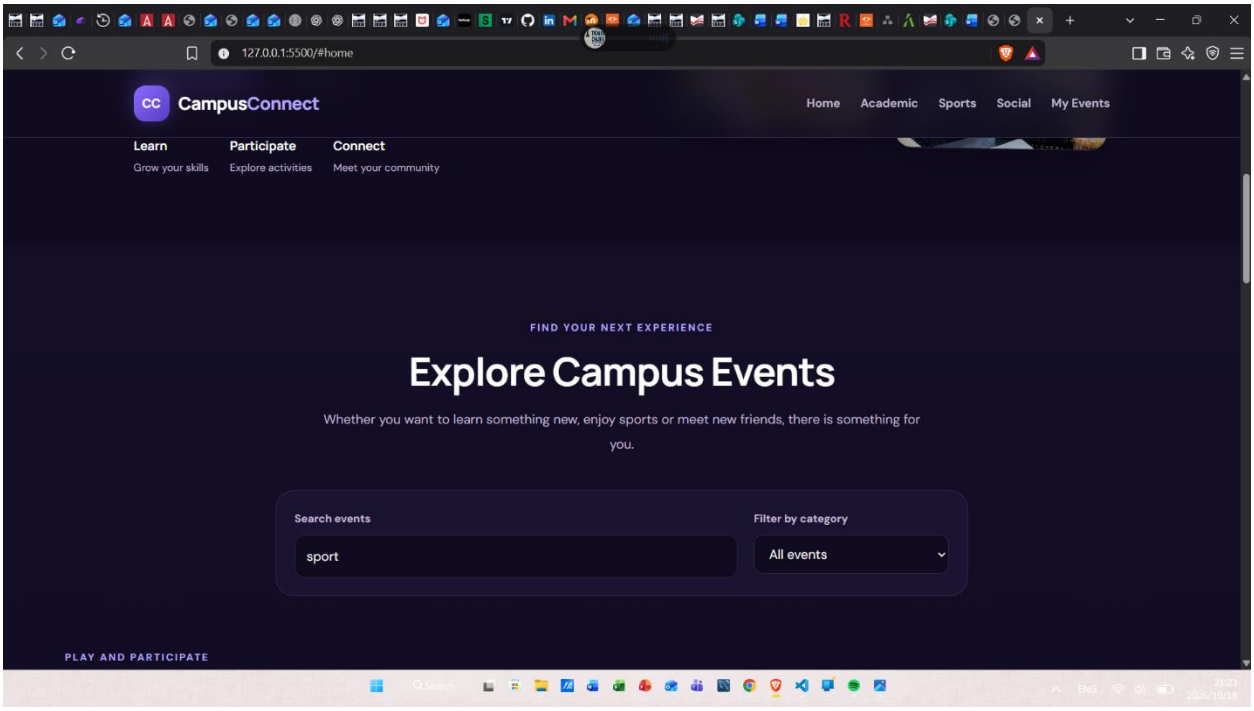
4. Robust

Website content should be compatible with different browsers and assistive technologies. CampusConnect uses HTML, CSS, and JavaScript to provide its interface and interactions. Semantic HTML, appropriate labels, and accessible controls should be checked to improve compatibility with assistive technologies.

Accessibility Evaluation

The WCAG 2.0 principles provide a framework for evaluating the accessibility of CampusConnect. Further testing is necessary to confirm that the website meets the relevant success criteria. Testing should include keyboard navigation, color contrast, alternative image text, form controls, and popup accessibility.

Overall, considering WCAG 2.0 helps identify improvements that can make CampusConnect more inclusive and easier to use for a wider range of students.


127.0.0.1:5500/#home
CampusConnect
Home
Academic
Sports
Social
My Events
Learn
Grow your skills
Participate
Explore activities
Connect
Meet your community
FIND YOUR NEXT EXPERIENCE
Explore Campus Events
Whether you want to learn something new, enjoy sports or meet new friends, there is something for you.
Search events
sport
Filter by category
All events
PLAY AND PARTICIPATE

References

Nielsen, J. (1994). Usability engineering. Morgan Kaufmann.

Example in-text citation for Section 4.1:

The CampusConnect prototype considers usability by helping students find relevant events, view event information, and register with minimal effort. These features support usability goals such as learnability, efficiency, and user satisfaction (Nielsen, 1994).

2. Interaction design principles

Use this source for design principles, buttons, feedback, visibility, and how users interact with the website.

Norman, D. A. (2013). The design of everyday things: Revised and expanded edition. Basic Books.

Example in-text citation for Section 4.3:

The prototype applies design principles such as visibility, consistency, and feedback. For example, the View Details buttons help users recognize available actions, while registration messages inform them when an action has been completed (Norman, 2013).

3. Accessibility and WCAG 2.0

Use this official source for Section 4.7.

World Wide Web Consortium (W3C)

Web Content Accessibility Guidelines (WCAG) 2.0.

Official accessibility guidelines covering perceivable, operable, understandable, and robust content.

Read the WCAG 2.0 guidelines

Example in-text citation for Section 4.7:

The CampusConnect prototype considers accessibility by using readable text, familiar navigation labels, and clear feedback messages. These design choices relate to the WCAG 2.0 principles of perceivable, operable, understandable, and robust content. Further testing is needed to determine whether the prototype meets the relevant success criteria (W3C, 2008)Use the heading References at the end of your report, then list the sources alphabetically:

Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.

Norman, D. A. (2013). *The design of everyday things: Revised and expanded edition*. Basic Books.

World Wide Web Consortium (W3C). (2008). *Web Content Accessibility Guidelines (WCAG) 2.0*. https://www.w3.org/TR/WCAG20/